

APPENDIX

A. Complexity and Performance on Real Datasets

	#classes	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10
N1'	2	0.0652	0.0621	0.6024							
	3	0.0702	0.6938								
	4	0.1438	0.6971	0.6066	0.1473						
	5	0.1472	0.7148	0.6137	0.1522	0.1131					
	6	0.3018	0.7122	0.6257	0.2041	0.1072	0.402				
	7	0.3662	0.7148	0.6427	0.1918	0.125	0.635	0.7775			
	8	0.3411	0.7198	0.6222	0.6386	0.127	0.633	0.7887	0.4257		
	9	0.6589	0.7206	0.6285	0.6175	0.1289	0.636	0.7875	0.4248	0.731	
	10	0.6455	0.57	0.7206	0.6532	0.6238	0.134	0.674	0.8388	0.4324	0.742
	2	0.2085	0.1912								
N2'	3	0.2071	0.4555	0.3600							
	4	0.2724	0.4423	0.3616	0.1957						
	5	0.2722	0.4400	0.3545	0.2010	0.2084					
	6	0.3038	0.4435	0.3760	0.1978	0.2033	0.2949				
	7	0.2992	0.4577	0.3828	0.2069	0.2077	0.3556	0.4664			
	8	0.3054	0.4452	0.3742	0.3706	0.2097	0.3553	0.4775	0.3302		
	9	0.4186	0.4503	0.3784	0.3697	0.2107	0.3554	0.4940	0.3304	0.4526	
	10	0.4247	0.3593	0.4476	0.3716	0.3632	0.2197	0.3746	0.4995	0.3331	0.4582
	2	0.0268	0.0218								
	3	0.0343	0.4010	0.3580							
N3'	4	0.0627	0.3976	0.3743	0.0235						
	5	0.0753	0.3850	0.3609	0.0272	0.0437					
	6	0.1547	0.3985	0.3821	0.0494	0.0456	0.1830				
	7	0.2040	0.4262	0.3913	0.0446	0.0506	0.3780	0.4712			
	8	0.2065	0.4077	0.3764	0.2624	0.0516	0.3680	0.4812	0.2313		
	9	0.4131	0.4177	0.3919	0.2586	0.0615	0.3720	0.4962	0.2346	0.4360	
	10	0.4105	0.3700	0.4128	0.3792	0.2760	0.0576	0.4420	0.5338	0.2254	0.4670
	2	0.8971	0.8959								
	3	0.8715	0.8705	0.9452							
	4	0.8583	0.7599	0.8369	0.8902						
F3'	5	0.8600	0.7080	0.7802	0.8642	0.7867					
	6	0.8143	0.6708	0.7357	0.8406	0.7081	0.7741				
	7	0.7800	0.6408	0.7062	0.7923	0.6584	0.7764	0.8200			
	8	0.7732	0.6393	0.7231	0.7639	0.6507	0.7169	0.7540	0.8511		
	9	0.7839	0.6133	0.6947	0.7847	0.6644	0.7031	0.7393	0.8253	0.9170	
	10	0.7491	0.4808	0.6152	0.6945	0.7668	0.6870	0.7333	0.7283	0.8261	0.8913
	2	0.9030	0.7853								
	3	0.7770	0.7985	0.9360							
	4	0.7634	0.6072	0.6963	0.7730						
	5	0.7403	0.5330	0.6113	0.7309	0.6201					
F4'	6	0.6888	0.4501	0.5348	0.6275	0.4979	0.3314				
	7	0.6505	0.3875	0.4758	0.6240	0.4163	0.4095	0.4392			
	8	0.6662	0.4116	0.4983	0.5920	0.4273	0.3819	0.4125	0.6523		
	9	0.6817	0.3787	0.4629	0.6209	0.4223	0.3991	0.4324	0.6106	0.7519	
	10	0.6190	0.1663	0.3816	0.4526	0.5709	0.3836	0.4350	0.4656	0.5560	0.7149
	2	0.7923	0.7874								
	3	0.7818	0.7953	0.8232							
	4	0.7660	0.7328	0.7606	0.7816						
	5	0.7635	0.7182	0.7399	0.7780	0.7956					
	6	0.7520	0.7093	0.7306	0.7478	0.7838	0.6998				
IN'	7	0.7417	0.6939	0.7155	0.7390	0.7637	0.6891	0.7148			
	8	0.7408	0.7031	0.7262	0.7316	0.7590	0.6815	0.6995	0.7847		
	9	0.7408	0.6892	0.7117	0.7336	0.7550	0.6840	0.6957	0.7672	0.7419	
	10	0.7285	0.5849	0.6757	0.6997	0.7147	0.7400	0.6800	0.6852	0.7504	0.7239

TABLE I: Complexity of the characters dataset

	#classes	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10
N1	c1	0.0446	0.1746								
	c2	0.0493	0.794	0.5187							
	c3	0.107	0.8112	0.5613	0.2509						
	c4	0.1271	0.8451	0.5915	0.2475	0.1726					
	c5	0.2562	0.811	0.6051	0.3437	0.1826	0.463				
	c6	0.305	0.7878	0.5841	0.3496	0.1825	0.6734	0.7768			
	c7	0.3194	0.8187	0.6111	0.8114	0.1843	0.6667	0.7679	0.3896		
	c8	0.6205	0.8324	0.6408	0.7643	0.1868	0.6993	0.8123	0.376	0.7496	
	c9	0.6245	0.5545	0.8678	0.6595	0.7828	0.1863	0.7085	0.8536	0.4045	0.7593
	c10	0.0991	0.0979								
N2	c1	0.1867	0.3464								
	c2	0.1987	0.5289	0.3942							
	c3	0.2591	0.5333	0.4066	0.271						
	c4	0.2491	0.5444	0.4192	0.2854	0.256					
	c5	0.2859	0.553	0.4307	0.3108	0.2629	0.3434				
	c6	0.2906	0.5514	0.4458	0.3198	0.2634	0.4153	0.4938			
	c7	0.2895	0.5523	0.4367	0.5129	0.279	0.4044	0.4881	0.323		
	c8	0.402	0.5417	0.4348	0.4893	0.273	0.4263	0.4987	0.3317	0.451	
	c9	0.4098	0.3322	0.5501	0.4368	0.5059	0.263	0.4304	0.5158	0.342	0.458
	c10	0.0991	0.0979								
N3	c1	0.0134	0.584	0.3395							
	c2	0.0401	0.5884	0.33	0.0588						
	c3	0.0384	0.6514	0.4172	0.0713	0.0337					
	c4	0.1294	0.5662	0.3767	0.1146	0.0516	0.2737				
	c5	0.1719	0.6115	0.4052	0.1638	0.0536	0.4682	0.5123			
	c6	0.1773	0.6225	0.4051	0.5016	0.0555	0.45	0.5036	0.2103		
	c7	0.3737	0.6101	0.3975	0.4584	0.0634	0.5088	0.532	0.1991	0.445	
	c8	0.3652	0.327	0.663	0.4473	0.4811	0.0575	0.4967	0.5543	0.2188	0.4658
	c9	0.8771	0.8704								
	c10	0.893	0.8403	0.8836							
F3	c1	0.8735	0.8528	0.7865	0.7923						
	c2	0.8712	0.8119	0.7419	0.774	0.7263					
	c3	0.8216	0.824	0.7867	0.7732	0.7444	0.567				
	c4	0.7901	0.8129	0.7904	0.7635	0.7532	0.6041	0.644			
	c5	0.8166	0.7871	0.7641	0.7471	0.7248	0.5679	0.6025	0.8518		
	c6	0.8134	0.7609	0.7858	0.7641	0.7473	0.5758	0.6049	0.8231	0.7761	
	c7	0.7787	0.6281	0.75	0.7293	0.724	0.698	0.6166	0.621	0.8214	0.7639
	c8	0.8323	0.7867								
	c9	0.8324	0.7346	0.7647							
	c10	0.772	0.6224	0.5522	0.5032						
F4	c1	0.7203	0.5154	0.5547	0.5171	0.4878					
	c2	0.6872	0.5166	0.5444	0.4919	0.4006	0.2732				
	c3	0.6487	0.4289	0.5129	0.4207	0.3136	0.3395	0.3968			
	c4	0.6761	0.4408	0.5183	0.4449	0.3371	0.3359	0.3754	0.6081		
	c5	0.6904	0.4436	0.5361	0.4875	0.4012	0.3776	0.4016	0.5763	0.565	
	c6	0.6376	0.2594	0.3925	0.4781	0.4343	0.3541	0.3592	0.4048	0.5234	0.5281
	c7	0.7884	0.7745								
	c8	0.8016	0.7831	0.7969							
	c9	0.7746	0.7809	0.7366	0.724						
	c10	0.7708	0.7802	0.7489	0.7279	0.736	0.6984				
Input Noise	c1	0.7586	0.7658	0.7456	0.7126	0.7241	0.6984				
	c2	0.7445	0.7523	0.7418	0.7041	0.7226	0.6859	0.6795			
	c3	0.7549	0.7543	0.7458	0.6989	0.7204	0.6786	0.667	0.7768		
	c4	0.7512	0.7438	0.7498	0.702	0.7264	0.6829	0.6687	0.7589	0.6969	
	c5	0.7373	0.5874	0.7267	0.7233	0.6963	0.7113	0.6775	0.662	0.7406	0.6824

TABLE II: Complexity of the characters-imb dataset

	#classes	c1	c2	c3
N1'	2	0.8295	0.4544	
	3	0.9427	0.8225	0.5086
N2'	2	0.5302	0.4586	
	3	0.5709	0.5199	0.4699
N3'	2	0.495	0.4298	
	3	0.7533	0.5425	0.5018
F3'	2	0.9954	0.995	
	3	0.9885	0.9907	0.9902
F4'	2	0.9912	0.994	
	3	0.9655	0.9827	0.9736
IN'	2	0.0952	0.0952	
	3	0.0947	0.0951	0.0944

TABLE III: Complexity of the connect-4 dataset

	#classes	c1</
--	----------	------

	#classes	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10
XGB	2	0.9938	0.9677								
	3	0.9913	0.6810	0.8442							
	4	0.9880	0.6292	0.8188	0.9744						
	5	0.9830	0.6011	0.8059	0.9702	0.9833					
	6	0.9585	0.5902	0.8068	0.9811	0.9882	0.9270				
	7	0.9481	0.5949	0.8009	0.9827	0.9861	0.8231	0.8092			
	8	0.9404	0.5775	0.7944	0.7894	0.9832	0.8008	0.7717	0.9311		
	9	0.9123	0.5639	0.7825	0.7717	0.9882	0.7874	0.7838	0.9174	0.8934	
	10	0.8915	0.4981	0.7706	0.8048	0.9810	0.7779	0.7273	0.9300	0.8886	0.8323
RF	2	0.9934	0.9653								
	3	0.9900	0.4983	0.7574							
	4	0.9835	0.4835	0.7591	0.9478						
	5	0.9806	0.5325	0.7825	0.9559	0.9783					
	6	0.9493	0.4712	0.7616	0.9595	0.9852	0.9103				
	7	0.9461	0.4686	0.7586	0.9650	0.9832	0.7799	0.7686			
	8	0.9478	0.4909	0.7623	0.7536	0.9745	0.7598	0.7617	0.9181		
	9	0.8785	0.4801	0.7603	0.7207	0.9843	0.7592	0.7512	0.9090	0.8534	
	10	0.8745	0.4495	0.7606	0.7626	0.9792	0.7336	0.6903	0.9223	0.8512	0.8508
TabNet	2	0.9183	0.2989								
	3	0.9183	0.2989	0.6276							
	4	0.7996	0.2273	0.5688	0.3789						
	5	0.7791	0.2341	0.5343	0.1162	0.6989					
	6	0.7138	0.2402	0.3296	0.5093	0.6705	0.4719				
	7	0.649	0.1556	0.3386	0.3578	0.7414	0.3601	0.3032			
	8	0.4923	0.0783	0.0	0.0	0.6134	0.3403	0.4219	0.6079		
	9	0.5233	0.2476	0.4769	0.0441	0.7251	0.4465	0.1674	0.6766	0.3157	
	10	0.0211	0.3554	0.1599	0.451	0.0551	0.4266	0.3139	0.181	0.4986	0.1068
MLP	2	0.9921	0.9595								
	3	0.9892	0.4400	0.6983							
	4	0.9633	0.1885	0.7207	0.9069						
	5	0.9705	0.4077	0.6972	0.9270	0.9719					
	6	0.8869	0.2524	0.7056	0.9145	0.9755	0.7971				
	7	0.8388	0.2546	0.6841	0.9158	0.9757	0.5611	0.4826			
	8	0.8322	0.4341	0.6034	0.2618	0.9656	0.5026	0.4880	0.8246		
	9	0.6192	0.2822	0.7190	0.1903	0.9710	0.4556	0.5015	0.8094	0.4458	
	10	0.5766	0.1999	0.6732	0.1578	0.9530	0.4035	0.4289	0.8216	0.4739	0.6530

TABLE VI: Performance (F1-Score) of the characters-imb dataset

	#classes	c1	c2	c3
XGB	2	0.5878	0.3742	
	3	0.8360	0.6167	0.4706
RF	2	0.7994	0.5831	
	3	0.6076	0.4137	0.47585
SVM	2	0.8154	0.57487	
	3	0.6322	0.4122	0.5368
LGB	2	0.8156	0.622	
	3	0.4497	0.404	0.1861
TabNet	2	0.6015	0.2204	
	3	0.8266	0.5615	0.2718
MLP	2	0.6125	0.2153	
	3	0.8578	0.5422	0.2887
TabPFN	2	0.8253	0.6366	
	3	0.6469	0.492	0.5578
TabTrans	2	0.8178	0.6166	
	3	0.6365	0.4637	0.5206

TABLE VII: Performance (F1-Score) of the cmc dataset

	#classes	c1	c2	c3
XGB	2	0.62941	0.8785	
	3	0.2391	0.7788	0.9038
RF	2	0.5119	0.8645	
	3	0.19324	0.7361	0.8908
SVM	2	0.54493	0.8608	
	3	0.2408	0.7488	0.8941
LGB	2	0.6472	0.8822	
	3	0.0488	0.6855	0.8902
TabNet	2	0.0527	0.8371	
	3	0.0019	0.5919	0.8421
MLP	2	0.5646	0.852	
	3	0.2201	0.7484	0.8955
TabPFN	2	0.6264	0.8804	
	3	0.2887	0.7994	0.9153
TabTrans	2	0.5751	0.8397	
	3	0.3115	0.7554	0.8993

TABLE VIII: Performance (F1-Score) of the connect-4 dataset